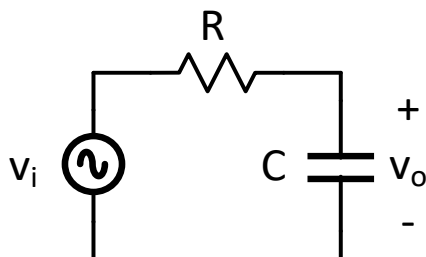


LAB 3 – FIRST-ORDER FILTERS

Introduction

In this lab, you will calculate and measure the frequency response of several different types of first-order filters. You will then generate Bode plots to communicate this information. For the first two circuits, we will perform these measurements manually. For the rest of the lab, however, we will use the Loopinator to automate this process.

Experiment 1. RC Low-Pass Filter



The low-pass filter shown above uses the following component values.

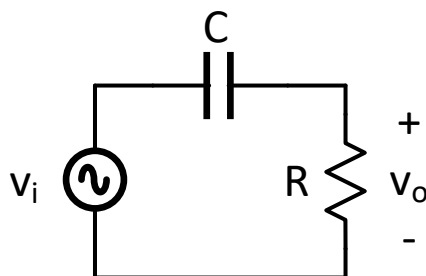
Component	Value
R	$1\text{ k}\Omega$
C	$0.1\text{ }\mu\text{F}$

1. Compute the transfer function $H(s) = v_o/v_i$ of the circuit above. Calculate the expected cutoff frequency and the expected magnitude of the transfer function at DC.
 - a. For more accurate results you can measure the resistor and capacitor values.
2. Build the circuit. The input is a sinusoid with an initial frequency of 100 Hz and an amplitude of $2.5\text{ V}_{\text{RMS}}$.
3. Measure the frequency response of the filter over the range of 100 Hz to 100 kHz. Measure at five points in each decade of frequency. It is suggested that you use the following distribution: 100, 150, 250, 400, 650; 1000, 1500, 2500, 4000, 6500, etc. This will give a reasonably smooth distribution of points within each decade.
 - a. At each frequency, use the voltmeter to measure the magnitude of the source voltage and the voltage drop across the capacitor. You should measure the source voltage, v_i , for every frequency.
 - b. At each frequency, use the oscilloscope to measure the phase difference between the voltage drop across the capacitor and the source voltage. Phase can be measured directly using the Measurement function on the scope. It can also be measured using the dime delay technique described in the tutorial. *If you use the*

time delay method, you will need to first turn on waveform acquisition averaging to get a reliable result.

4. Use the Excel spreadsheet provided on Canvas to plot the magnitude (in dB) and phase of both your measured and calculated frequency response. Include the result in your lab report.

Experiment 2. RC High-Pass Filter



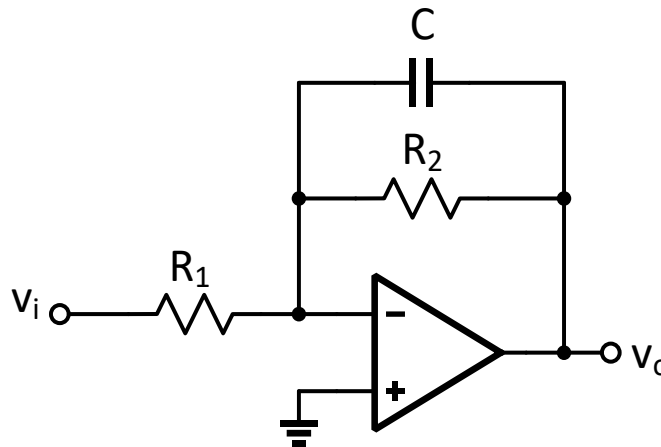
The high-pass filter shown above uses the following component values.

Component	Value
R	$1.5\text{ k}\Omega$
C	$0.1\text{ }\mu\text{F}$

1. Compute the transfer function $H(s) = v_o/v_i$ of the circuit above. Calculate the expected cutoff frequency and the expected magnitude of the transfer function at $\omega \rightarrow \infty$.
 - a. For more accurate results you can measure the resistor and capacitor values.
 - b. All work should be included in the appendix of the lab report.
2. Build the circuit. The input is a sinusoid with an initial frequency of 100 Hz and an amplitude of $2.5\text{ V}_{\text{RMS}}$.
3. Measure the frequency response of the filter over the range of 100 Hz to 100 kHz. Measure at five points in each decade of frequency. It is suggested that you use the following distribution: 100, 150, 250, 400, 650 – 1000, 1500, 2500, 4000, 6500, etc. This will give a reasonably smooth distribution of points within each decade.
 - a. At each frequency, use the voltmeter to measure the magnitude of the source voltage and the voltage drop across the capacitor. You should measure the source voltage, v_i , for every frequency.
 - b. At each frequency, use the oscilloscope to measure the phase difference between the voltage drop across the capacitor and the source voltage. Phase can be measured directly using the Measurement function on the scope. It can also be measured using the time delay technique described in the tutorial. *If you use the time delay method, you will need to first turn on waveform acquisition averaging to get a reliable result.*

4. Use the Excel spreadsheet provided on Canvas to plot the magnitude (in dB) and phase of both your measured and calculated frequency response. Include the result in your lab report.

Experiment 3. Active Low-Pass Filter

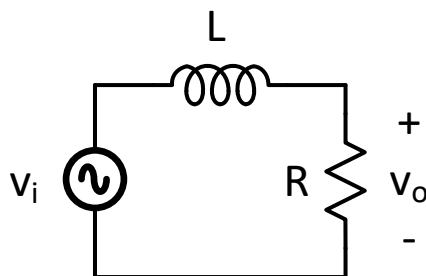


The active low-pass filter shown above uses the LMC660 op amp with ± 8 V supplies. All other component values are given in the table below.

Component	Value
R_1	$1\text{ k}\Omega$
R_2	$10\text{ k}\Omega$
C_1	10 nF

1. Compute the transfer function $H(s) = v_o/v_i$ of the circuit above. Calculate the expected cutoff frequency and the expected magnitude of the transfer function at DC.
 - a. For more accurate results you can measure the resistor and capacitor values.
 - b. All work should be included in the appendix of the lab report.
2. Build the circuit. The input is a sinusoid with an initial frequency of 100 Hz and an amplitude of $0.25\text{ V}_{\text{RMS}}$.
3. Use the Loopinator to measure the frequency response of the filter from 100 Hz to 100 kHz with at least 15 points per decade. Use an input voltage of $0.25\text{ V}_{\text{RMS}}$.
4. Create a Bode plot showing both your measured and calculated values on the same plots. Use dB for the magnitude and degrees for the phase. Include these plots in your lab report.

Experiment 4. RL Low-Pass Filter



The active low-pass filter shown above uses the component values shown in the table below.

Component	Value
L_1	60 mH
R_1	680 Ω

1. Compute the transfer function $H(s) = v_o/v_i$ of the circuit above. Calculate the expected cutoff frequency and the expected magnitude of the transfer function at DC.
 - a. All work should be included in the appendix of the lab report.
2. Build the circuit. You should use four 15 mH inductors in series in order to realize the required 60 mH. The input is a sinusoid with an amplitude of 2.5 V_{RMS}.

We do not require a complete frequency response of this circuit. Instead, we will estimate the low-frequency gain and cutoff frequency from measurements.

3. Set the frequency at least two decades below the expected cutoff frequency (somewhere around 10 Hz is sufficient).
4. Use the voltmeter to measure the source voltage, v_i , and the voltage drop across the resistor, v_o . Use these two measurements to calculate the low-frequency magnitude of the transfer function. Record the values of v_i and v_o and include them in your lab report.
5. Divide the low-frequency magnitude calculated above by $\sqrt{2}$. In other words, we are computing the expected magnitude at the cutoff frequency, $|H(0)|/\sqrt{2}$.
6. Now, keeping the source amplitude constant, increase the frequency until v_o reduces to match your calculated value. The frequency at which this occurs is the measured cutoff frequency. Again, include all these values in your lab report. Record the values of v_i and v_o at this frequency and include them in your lab report.

You will probably find that your measured low-frequency gain and the measured cutoff frequency are significantly different from your calculated values. This is due to nonideal losses in the inductor.

7. Measure the inductance and series resistance of all four inductors using the LCR meter.
 - a. Set the mode to $L_s - R_{dc}$ and set the measurement frequency as close to your calculated cutoff frequency as possible.
 - b. Include these values in your lab report. When reporting these values, report the sum all of the inductances as Total Inductance and the sum of all of the resistances as Total DCR.
8. Recalculate the transfer function but replace the inductor with a more accurate model of an inductor in series with a resistor, called R_{DCR} . The output voltage is still taken across the $680\ \Omega$ resistor.
 - a. The values of the inductor and newly added R_{DCR} will come from the values measured using the LCR meter. In addition, you should also use the measured value of R (the $680\ \Omega$ resistor) as well.
9. Compare the new calculations to your measured values. They should be much closer.

Experiment 5. Design of an Active High-Pass Filter

Design, build, and test an active first-order high-pass filter. Use the LMC660 op amp with $\pm 8\text{ V}$ supplies along with resistors and capacitors from your lab kit. Inductors cannot be used for this design. Your filter should meet the following specifications:

Parameter	Value
Cutoff Frequency	1 kHz $\pm 10\%$
Pass-Band Gain	-5 V/V $\pm 10\%$

1. Use LTSpice to perform AC simulations of your proposed design from 100 Hz to 100 kHz. Include a screenshot of the simulation results in your lab report.
2. Take a screenshot of the oscilloscope showing the input and output waveforms together at the cutoff frequency.
3. Use the Loopinator to measure the full frequency response of your filter from 100 Hz to 10 kHz using 15 points per decade. Include a plot of the measured frequency response (magnitude and phase) in your lab report.